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The Impact of Laterality and Prosthesis on Performance Among Para Surfers: A Cross-Sectional Study

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ABSTRACT

Purpose: The purpose of this study was to investigate performance differences in competitive adaptive/para surfers with upper and/or lower limb amputations, congenital limb impairments, or dysmelia who surf in the standing position. **Methods:** This cross-sectional study explored the impact of wave direction; laterality, or surf stance, such as goofy (right foot forward) versus regular (left foot forward); and use of prosthesis on performance using judged wave scores among competitive surfers from 2019 to 2022. Archived records of classified surfers with eligible impairment of limb deficiency were reviewed. Data was divided into surfing profile and surfing performance. **Results:** Participants ($n = 86$; male = 64; female = 22) from 22 nations met inclusion criteria. No significant difference was found in wave direction, laterality, or prosthesis use; however, more frontside waves were ridden. Significant differences were found between male (single-wave score 4.62/10; total wave score 9.25/20) versus female surfers (single-wave score 3.44/10; total-wave score 6.82/20) for average single wave score ($t(49,490) = 2.989$, $p = .004$, two-tailed; $d = .647$) or average total wave score ($t(48,768) = 2.585$, $p = .003$, two-tailed; $d = 0.670$). **Conclusion:** Although limited to sample size, adaptive/para surfing performance may be more influenced by training and skill acquisition than laterality or use of prosthesis.

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KEYWORDS

Classification; para surfing; paralympics; prosthesis

Lateral preferences of how humans use the right or left sides of their bodies, such as handedness or standing, are shaped by behavior and dependent upon the activity or sport (Furley et al., 2018). How one prefers to stand on a board in action sports, such as skateboarding, snowboarding, or surfing, is described as goofy footed, with the right foot forward, and regular footed, with the left foot forward (Borgonovo-Santos et al., 2021; Furley et al., 2018). Sport stance is known as laterality. Furley et al. (2018) found laterality and wave direction influenced surfing performance in professional able-bodied surfers. This study sparked a curiosity to investigate the influence of lateral preferences and wave direction in professional adaptive/para surfers.

Adaptive surfing ranges from participating in surf therapy, learning how to surf, participating in local surf contests, and eventually competing as an adaptive professional surfer on the Adaptive World Tour organized by the Association of Adaptive Surfing Professionals (AASP) (<https://adaptivesurfingprofessionals.com>) and/or competing for one's nation as a para surfer at the World Para Surfing Championships organized by the International Surfing Association (ISA) (<https://isasurf.org>) (AASP; ISA; Clapham et al., 2014). The terms *adaptive surfer* and *para surfer* will be combined as *adaptive/para surfer* denoting the highly competitive level of the athletes in this study who have been officially classified and deemed eligible to compete on both platforms. Professional adaptive/para surfers are individuals with permanent physical or visual impairments who are competing at elite levels in Paralympic-style competitions (Johnson & David, 2022).

Where able-bodied surfers are separated by gender and surf style, such as short boarding or longboarding, Johnson and David (2022) described how adaptive/para surfers are divided into different sport classes based on their surf style and severity of their permanent physical or visual impairments. The current classification structure includes the following nine sport classes differing from underlying health conditions, eligible impairments, minimal impairment criteria, and surf position: Para Surfing (PS) Stand 1, 2, and 3; PS Kneel/Upright; PS Sit; PS Prone 1 and 2; and PS Vision Impairment 1 and 2 (Johnson & David, 2022). The classification system is vital because if the athletes are not classified accurately, some may have an unfair competitive advantage over others, and the outcome could be predictable (International Paralympic Committee [IPC], 2020). The present research analyzed only the PS Stand 1, 2, and 3 sport classes because these athletes were the most appropriate samples as they represent adaptive/para surfers who engage in similar activities and perform surfing in the standing position as in the study by Furley et al. (2018). PS Stand 1 is for surfers who have an upper limb amputation, congenital limb impairment, or dysmelia. PS Stand 2 is for surfers with a transtibial amputation (TTA) or congenital equivalent limb impairment. PS Stand 3 is for surfers who have a transfemoral amputation (TFA) or both lower extremities with TTA or congenital equivalent limb impairment (Johnson & David, 2022).

Performance in able-bodied sports has been found to be affected by the athletes' stance (Baker et al., 2013; Furley et al., 2018; Loffing & Hagemann, 2016). Surfers stand facing

perpendicular on the surfboard when riding waves with either the left foot forward or the right foot forward (Furley et al., 2018). Like surfing, athletes who snowboard or skateboard also have a lateral preference. Staniszewski et al. (2016) found in snowboarding, although performance was affected by lateral preference, it was more affected by balance ability and postural control.

Inquiries on combat sports, such as judo, boxing, and fencing, found lateral preference had a significant performance factor based on a *negative frequency-dependent* selection. This factor means that because there is less representation for left-dominant athletes, these athletes were more likely to be successful because right-dominant athletes were less well adapted for competing against left-dominant athletes (Ardá Suárez et al., 2016). Laterality differences were also discovered among athletes in non-combat sports like cricket where the batsmen were seven times more likely to use a reversed stance, indicated by the position of the dominant hand on top of the handle, which gave them a significant performance advantage (Mann et al., 2016). Additionally, a significant performance advantage was found in throwing right-handed and batting left-handed in professional and major league baseball players (Mann et al., 2016).

Zouhal et al. (2018) found laterality influenced agility, better performance, and faster movement time if the soccer athlete turned on their non-dominant leg and pushed off with their dominant leg when performing a common 180° rotation while running (Zouhal et al., 2018). Fletcher and Long (2013) found left-sided male soccer players had significantly better dominant leg balance when compared to right-sided players. On the contrary, elite female Czech youth alpine skiers had no correlation between the dominant and non-dominant legs regarding skiing balance ability (Turkova et al., 2021).

Although an increasing body of research has provided evidence for a relationship between laterality and performance in able-bodied sports, only isolated studies were found on Paralympic-level athletes and effects of laterality on performance in adaptive sports. Prosthetic technology and use may influence performance with running direction and the side of amputation in Paralympic sprinters (Funken et al., 2019). Tuakli-Wosornu et al. (2021) investigated prosthetic blade technology used over a 20-year period and found competitive race speed advantages in the Paralympians with bilateral lower limb amputations compared to Paralympians with unilateral lower limb amputations.

Residual limb lengths and use of the prosthesis may influence surfing performance. Surfers with TFA must select one of the three surf stance for competing: standing with prosthesis, kneeling without prosthesis, or a deep squatted position without a prosthesis. The trend is that surfers with shorter residual femurs preferred to surf in a kneeling or deep squatted position without a prosthesis, and surfers with longer residual femurs preferred to surf in a standing position with a prosthesis. Johnson et al. (2022) found comparable performance results despite the surf stance and that overall surfers with TFA outperformed surfers with TTA based on judged wave scores. Previous experience and time spent surfing

contributed to better surfing skills and increased participation (Cruz et al., 2021).

Environmental features may play a role in motor laterality. Participants racing in hurdling used their right leg as the lead hurdling leg at a running pace and changed their leading leg depending on the running speed, course shape, or with the directional change of the track (Iskra et al., 2019). Like able-bodied surfers, competitive adaptive/para surfers are dependent on the direction of the breaking wave. Scarfe et al. (2003) illustrated the direction of the breaking waves as left or right peeling waves and further explained that the direction that a wave breaks is the direction the surfer must go. Therefore, surfers must either go to the front side, facing the wave or backside, with their back to the wave, depending on the wave direction and their surf stance as shown in Figure 1.

The purpose of this study was to investigate performance differences in competitive adaptive/para surfers with upper and/or lower limb amputations, congenital limb impairments, or dysmelia who surf in the standing position. The impact of surf stance, wave direction, and use of prosthesis on surf performance was analyzed. The first hypothesis was that adaptive/para surfers facing the wave would have a better performance as they would be able to see and gather better wave information more efficiently than surfing backside (Furley et al., 2018). The second hypothesis was that a goofy or regular stance would not affect performance like able-bodied professional surfers found by Furley et al. (2018). The third hypothesis was that surfers who used a prosthesis on their front leg would outperform surfers who used a prosthesis on the back foot as turning maneuvers originate from the back foot (Mendez-Villanueva et al., 2010). It was speculated that because the surfboard is controlled by the back foot and hips, it might be harder to maneuver and control with a prosthesis. The fourth hypothesis was that surfers with TFA using a deep squatted position without a prosthesis would outperform surfers with TFA who stood using a prosthesis, because those without a prosthesis would have less compensatory motions to pop up to the wave riding position. Lastly, it was hypothesized that laterality preference would not impact surf performance for those with an upper limb amputation, congenital limb impairment, or dysmelia, rather the surf direction and if the surfer faces toward the wave will cause better performance due to available visual input.

Materials and methods

Participants

Archived data from classification records ($n = 96$) were reviewed for surfers classified into the PS Stand 1, PS Stand 2, and PS Stand 3 sport classes with the eligible impairments of limb deficiency described as TFA, TTA, congenital limb impairment, dysmelia, or brachial plexus injury. The present study examined professional adaptive/para surfers classified into the PS Stand 1, 2, and 3 sport classes. The inclusion criteria were that all eligible participants must be over 18 years old, classified into a para surfing standing sport class, and had competed in at least one of the international AASP or ISA surfing competitions

Right Breaking Waves	Left Breaking Waves
	
Regular footed-front side, surfer with below the knee prosthesis	Regular footed-backside, surfer with below the knee prosthesis
	
Goofy footed-backside, surfer with back arm impairment	Goofy footed-frontside, surfer with front arm impairment

Figure 1. Surf stance, wave direction, and limb impairment. *Note.* Surfing stance, wave direction, and impairments or prosthesis noted. The left column represents right-breaking waves, and the right panels represent left-breaking waves. (Modified from Furley et al., 2018). Images used with permission from ISA and Stoke for Life Foundation.

from 2019 to 2022. Data with at least one rating of the surfing score in any of the competitions were used. If the participants competed in more than one competition, the best score was used for the correlational analysis. Data were excluded if the participants were under 18 years old, were not classified into a para surfing standing sport class, and/or did not compete in any of the competitions from 2019 to 2022. Even though most locations have a consistent wave direction, the wave scores were cross-referenced with competition video footage to ensure accuracy for wave direction. For example, Queens Break in Hawaii has a consistent right; however, during many of the heats, there was a cross swell, and the competitors were able to go left on the break. All participants provided signed informed consent under approved ethical guidelines from the Institutional Review

Board with Protocol IRB# 0731-019A. STROBE guidelines were followed for reporting of cross-sectional studies.

Study design

The first part of this study involved designing a surfer profile from retrieved data to learn features about each surfer and demographics. The second part of this study was to compare surf stance laterality and wave direction to the recorded top two best waves and contest rankings.

Study part 1 adaptive/para surfer profiles

The following information was retrieved and de-identified from classification forms: gender, age, nationality, diagnosis, use of prosthesis, years surfing, quantity of competitions, surf

stance, and classified sport class. Participants were grouped by classification sports classes of PS Stand 1 with upper limb involvement; PS Stand 2 with below the knee lower limb involvement; PS Stand 3 with above the knee lower limb involvement; and gender (male/female). Surfer information is routinely collected at the time of the classification. Data from classification records for this study were retrieved during September 2022 through December 2022.

If a participant answered a question on the classification form in a number range, the average between the two numbers was used for their data. For example, if the question was, “How many times do you go surfing in a month?” and the participant responded with “15–18 times,” 16.5 was reported for the data. The same calculation was used if the participant responded with a number range for the question, “How long is your average surf session in hours?.” When asked, “How many surf sessions do you surf in a month?” and a surfer reported “every day” then the number 30 was used to analyze data. If surf style, affected side, surfboard style, or prosthetic leg was not reported in the participant profile, video footage on the contest websites and social media platforms was used and reviewed by researchers [MJ & AI] to identify. Shortboard length was identified as 7'11" and shorter and longboard length was identified as 8.0 and longer (Scarfe et al., 2003). Marked foam surfboards were designated in their own category. Age was calculated with the current year (2023) and birth year. Hand dominance was not considered due to lack of reported data.

Study part 2 adaptive/para surfing performance

Surf contests occur in ocean conditions with varying wave types and sizes plus fluctuating weather and tides, which influence surfing performance (Farley et al., 2012, 2017; Mendez-Villanueva & Bishop, 2005). Surfers strive to catch and ride the biggest or best waves and perform complex maneuvers to outscore their competitors (Farley et al., 2012,

2017; Mendez-Villanueva & Bishop, 2005). Surf contests for women, men, and adaptive/para surfers alike utilize tournament-style rounds, with each round made up of 4-surfer competition heats. Surfers progress through rounds, quarterfinals, and semifinals to advance to the finals. The final heat determines the overall contest winners.

Competitive surfing success is determined by judging criteria that evaluate the surfer's ability to ride a surfboard while executing innovative and athletic maneuvers in the most critical parts of the breaking wave. Certified judges score surfers on a 10-point scale, with 1 being a poor ridden wave and 10 being an excellent ridden wave. In one heat, each surfer may catch up to 10 waves; however, only the top two highest-scoring waves are added together for their total score out of 20 points. Participants' two highest scores from the judges' score sheets from the competitions were collected to determine overall surfing performance.

The seven contests from 2019 to 2022 occurred at different locations in California and Hawaii. Overall contest rankings were pulled from event websites (AASP, ISA, and Stact App). Each surf spot was recorded for location, year, organization, wave break type, ocean floor terrain, wave direction, and winning stance per gender as outlined in Table 1.

Two researchers [MJ and AI] reviewed input data for the surfers' profiles, laterality, wave scores, and contest rankings multiple times to report data as accurately as possible. If there were any discrepancies, the reviewers triple checked to ensure accuracy.

Performance was determined by the top two best judged wave scores and overall contest ranking for each competition. Performance data were retrieved from contest websites: ISA; AASP; and the Stact App (Association of Adaptive Surfing Professionals [AASP], 2022; International Surfing Association [ISA], 2022b, 2022c, 2022d; Stact App, 2022). In addition, the final judged wave score sheets and rankings were provided by

Table 1. Chronological order of event sites and organization, location information, and winning stance.

Surf Spot Year, Association	Surf location	Break type	Bottom	Wave direction	Winner stance Men	Winner stance Women
Oceanside Pier Southside 2019, AASP	Oceanside, California, USA	Beach break	Sand	Right and left	PSS1-regular PSS2-goofy PSS3-regular	PSS1-goofy PSS2-regular PSS3-regular
La Jolla Shores 2020, WPSC	La Jolla, California, USA	Beach break	Sand	Right and left	PSS1-goofy PSS2-regular PSS3-goofy	PSS1-goofy PSS2-regular PSS3-NS
Oceanside Pier Northside 2021, AASP	Oceanside, California, USA	Beach break	Sand	Right and left	PSS1-goofy PSS2-goofy PSS3-goofy	PSS1-goofy PSS2-regular PSS3-NS
Pismo Pier Southside 2021, WPSC	Pismo, California, USA	Beach break	Sand	Right and left	PSS1-goofy PSS2-regular PSS3-goofy	PSS1-goofy PSS2-regular PSS3-NS
Queens 2022, AASP	Honolulu, Hawaii, USA	Point break	Reef	Right with an occasional left	PSS1-goofy PSS2-regular PSS3-goofy	PSS1-goofy PSS2-regular PSS3-NS
Oceanside Pier Northside 2022, AASP	Oceanside, California, USA	Beach break	Sand	Right and left	PSS1-goofy PSS2-regular PSS3-goofy	PSS1-goofy PSS2-regular PSS3-NS
Pismo Pier Southside 2022, WPSC	Pismo, California, USA	Beach break	Sand	Right and left	PSS1-regular PSS2-regular PSS3-goofy	PSS1-goofy PSS2-regular PSS3-regular

Note. AASP = Association of Adaptive Surfing Professionals; ISA = International Surfing Association; PSS1 = Para Surfing Stand 1, surfers with upper limb impairments; PSS2 = Para Surfing Stand 2, surfers with transtibial amputation or equivalent limb impairments; PSS3 = Para Surfing Stand 3, surfers with transfemoral amputation or equivalent limb impairments; NS = No Surfer competed in the sport class. Table modified with permission from Furley et al. (2018).

competition directors. All judges earned certification from the ISA Judging and Officiating Course (International Surfing Association [ISA], 2022a). Judging wave scores were independent of classification and no communication occurred between the classifiers, the judges, or the researchers. Ranking was based on the accumulation of scores per surfer.

Statistical analyses

The first two hypotheses regarding performance differences as a function of surf stance and wave direction were tested with factorial ANOVAs and dependent and independent t-tests. Similarly, the third hypothesis that surfers who used a prosthesis on their front leg would outperform surfers who used a prosthesis on their back leg were tested with independent t-tests. Hypotheses 4 (surfers with TFA using a deep squatted position without a prosthesis would outperform surfers with TFA who stood using a prosthesis) and 5 (laterality preference would not impact surf performance for those with an upper limb amputation, congenital limb impairment, or dysmelia) were tested with a series of t-tests. The dependent variables indicative of wave performance were the judged scores and resulting rankings. We denote results as significant if p values fall below .05. Sensitivity analyses conducted with g^* Power 3.1.9.7 for the different statistical tests indicated that tests were only able to detect moderate to large statistical effects.

Results

Study part 1: surfer profiles

A cross-sectional comparative study design analyzed archived data from seven international adaptive/para surfing competitions and classification records. Of the 96 reviewed classification records, 86 participants (male = 64; female = 22) met inclusion criteria and participated in this study representing PS Stand 1 ($n = 26$); PS Stand 2 ($n = 41$); and PS Stand 3 ($n = 19$). Ten records were excluded due to exclusion criteria for being under the age of 18 years old ($n = 7$) or surfers with short stature who did not have leg amputations ($n = 3$). The age range was 18–62 years old with the mean age of 39.05 ± 21.5 . The participants came from 22 different nations within five regions of the world: Africa ($n = 3$); Americas ($n = 44$); Asia ($n = 7$); Europe ($n = 30$); and Oceania ($n = 2$). The average surfing experience was 11.3 years; average surf sessions a month was 14.2 times lasting approximately 2 h; and the average number of surf contests was 9.6.

Study part 2: laterality and surfing performance

Analysis results on how surfing facing the wave compared to backside to the wave influenced surfing performance in competitive adaptive/para surfers found no effect; however, more front facing waves were ridden in the testing period as seen on video footage. Two-factor univariate ANOVA on the average scores of surfers with the between-subject factors stance (regular vs. goofy) and wave direction (left vs. right) revealed no

main effect for stance, $F(1, 37) = 0.401$, $p = .530$, $\eta^2_p = .011$; no main effect for wave direction, $F(1, 37) = 1.797$, $p = .188$, $\eta^2_p = .046$; and no interaction between stance and wave direction, $F(1, 37) = 0.450$, $p = .507$, $\eta^2_p = .012$. This pattern of results was further supported by non-significant dependent t-test comparing the average frontside (4.752 points) vs. backside score (4.980 points; $t(38) = -0.841$, $p = .406$, two-tailed; $d = -0.135$) and average scores for right waves (5.058 points) vs. left waves (4.673 points; $t(38) = 1.449$, $p = .078$, two-tailed; $d = 0.232$). However, there was a significant difference between the total number of frontside (2.77/10) vs. backside (1.88/10) waves that went into the top two waves contributing to the two-wave-total score during the analyzed time period (7 contests) $t(84) = 3.312$, $p = .001$, two-tailed; $d = 0.359$).

Each of the seven events had six possible winners, three male and three female champions. For PS Stand 1 winner stance: goofy ($n = 12$) and regular ($n = 2$) stance. For PS Stand 2 winner stance: goofy ($n = 2$) and regular stance ($n = 12$). For PS Stand 3 winner stance: goofy ($n = 6$), regular stance ($n = 3$), and 5 events did not have women surfers (see Table 1). This study found most of the participants had a goofy stance ($n = 46$; 55%) over a regular stance ($n = 39$; 45%). Overall, no significant differences were found in surfing performance with surfers using a goofy compared to a regular stance for each sport class. Goofy surfers (rank 5.556; single wave score 4.065; total wave score 8.147) vs. regular surfers (rank 4.901; single wave score 4.590 total wave score 9.181) and no significant difference including average ranking ($t(76) = 1.045$, $p = .299$, two-tailed; $d = 0.237$), average single wave score ($t(75) = -1.221$, $p = .226$, two-tailed; $d = -.279$) and average total wave score ($t(67.678) = -1.200$, $p = .234$, two-tailed; $d = -0.277$). However, there were significant gender differences in performance. There were significant differences between male (single wave score 4.617; total wave score 9.246) vs. female surfers (single wave score 3.435; total wave score 6.821) for average single ($t(49.490) = 2.989$, $p = .004$, two-tailed; $d = .647$) or average total wave score ($t(48.768) = 2.585$, $p = .003$, two-tailed; $d = 0.670$). Stance, sport class, and gender compared to judged wave scores are depicted in Figure 2.

The affected leg was reported with front ($n = 18$, 29%); back ($n = 30$, 48%), and both ($n = 13$, 21%). For the 46 goofy-footed participants, front leg was affected ($n = 6$, 13%); back leg was affected ($n = 18$, 39%); and both legs were affected ($n = 8$, 17%). For the 39 regular-footed participants, front leg was affected ($n = 12$, 31%); back leg was affected ($n = 13$, 33%); and both legs were affected ($n = 4$, 10%).

There were no significant differences between surfers with a TFA or a TTA who wore a prosthesis on their front foot compared to their back foot on their average wave score ($t(40) = 0.495$, $p = .623$, two-tailed; $d = .159$), average rank ($t(41) = 0.279$, $p = .782$, two-tailed; $d = .088$), and average total score ($t(41) = .498$, $p = .621$, two-tailed; $d = .157$). Further, there were no differences on the average frontside scores ($t(32) = -0.295$, $p = .770$, two-tailed; $d = -.104$) or backside scores ($t(26) = 0.621$, $p = .782$, two-tailed; $d = .088$) for the single waves as shown in Figure 3.

Surfers with TTA must surf standing with prosthesis; however, surfers with TFA have the option to stand with a prosthesis or surf without one in competition. There were no significant differences between surfers who wore

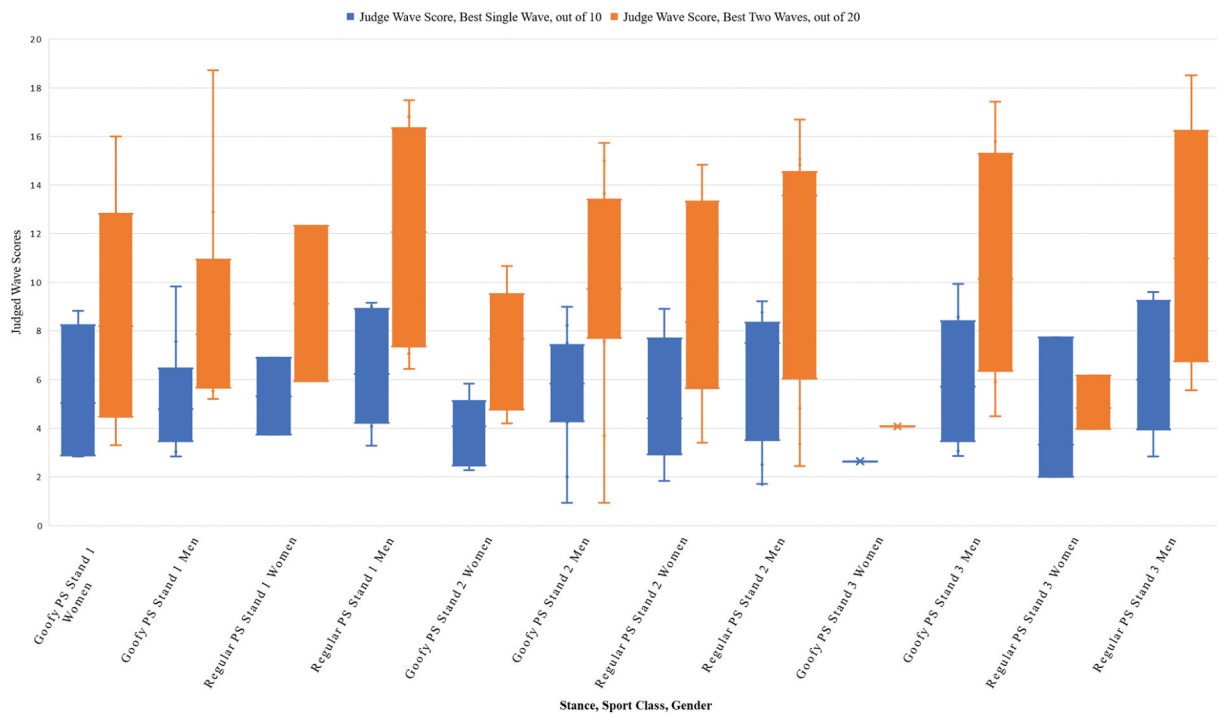


Figure 2. Stance, sport class, and gender compared to judged wave scores.

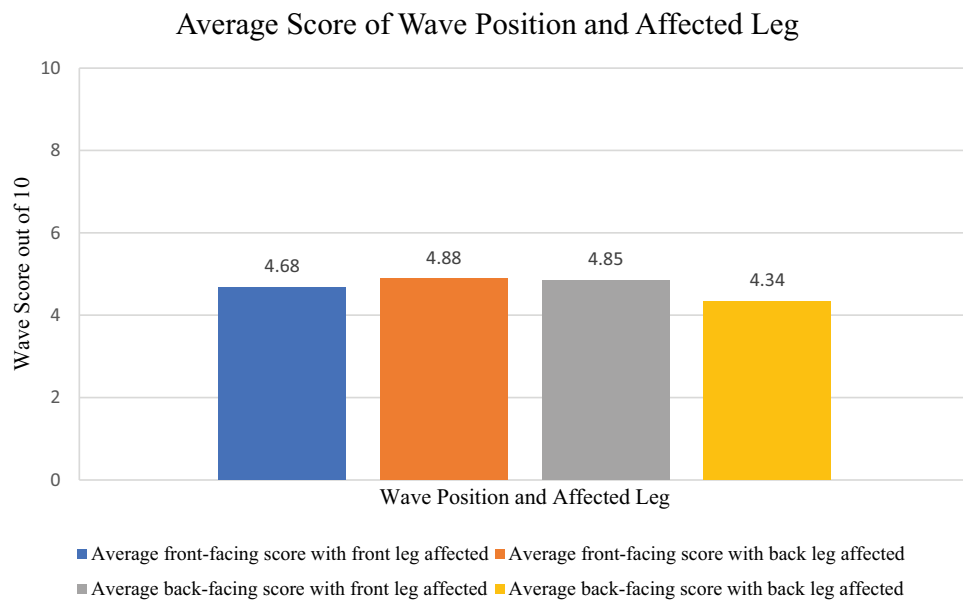


Figure 3. Average wave scores of wave position and affected leg.

a prosthesis during competition compared to surfers who did not wear a prosthesis on their leg with average wave score ($t(50) = -0.031, p = .975$, two-tailed; $d = -.009$), average rank ($t(51) = 0.678, p = .501$, two-tailed; $d = .200$), and average total score ($t(51) = .018, p = .986$, two-tailed; $d = .005$). Further, there were no significant differences on the average front-side scores ($t(29.279) = -0.380, p = .706$, two-tailed; $d = -.121$) or backside scores ($t(33) = -0.439, p = .663$, two-tailed; $d = -.160$) for the single wave scores.

When comparing surfers with TFA only, there were no significant differences between surfers who wore a prosthesis

compared to surfers who did not wear a prosthesis on their average wave score ($t(16) = -0.323, p = .751$, two-tailed; $d = -.151$), average rank ($t(16) = 0.738, p = .161$, two-tailed; $d = .200$), and average total score ($t(16) = -.355, p = .727$, two-tailed; $d = -.167$). Further, there were no significant differences among surfers with TFA on the average frontside scores ($t(14) = -0.658, p = .521$, two-tailed; $d = -.329$) or backside scores ($t(9) = -0.328, p = .750$, two-tailed; $d = -.199$) for the single wave scores as seen in Figure 4.

Hand dominance was not reported or captured in this study; therefore, surf stance was compared to front versus

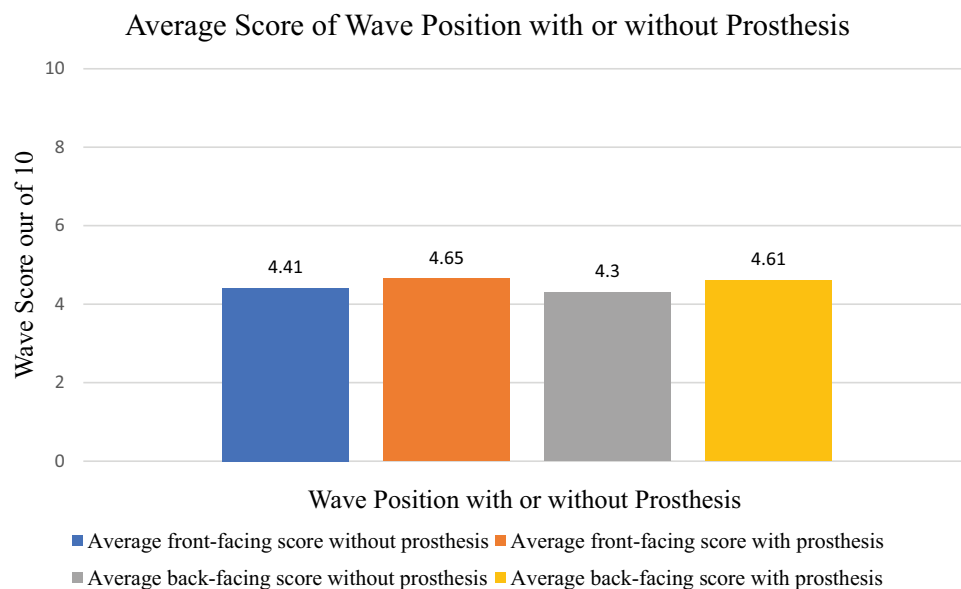


Figure 4. Average wave score using wave position and standing with or kneeling without prosthesis.

back upper limb affected. Results found a goofy stance ($n = 16$) had front arm ($n = 7$), back arm ($n = 7$), or both arms ($n = 2$) affected. Results found that a regular stance ($n = 12$) had front arm ($n = 5$), back arm ($n = 7$), or both arms ($n = 0$) affected. Regarding the last hypothesis on the arms, no significant differences were evident regarding stance on any performance measure (all $p > .14$). Additionally, no performance differences were evident depending on whether the back or the front arm was affected (all $p > .08$).

Discussion

This study analyzed the impact of wave direction, laterality, and use of prosthesis on surfing performance using judged wave scores and contest rankings. Based on the surfers' profiles, the participants in this study were experienced surfers with over 10 years of adaptive surfing experience and training.

Facing or backside to wave direction

The first hypothesis investigated performance with facing or back side to the wave direction. Although it seems plausible that surfing stance and wave direction interact in creating facilitative conditions for surfing when surfing frontside as compared to backside in adaptive/para surfers, this theory did not show in the data. We reasoned that surfing frontside would be beneficial for picking up relevant visual information to guide the motor actions necessary for successful surfing. However, riding facing or backside to a peeling wave had no effect on adaptive/para surfing performance comparing average front versus backside wave riding scores for right and left wave directions which can also be found in able-bodied surfing studies (Farley et al., 2017; Forsyth et al., 2017, 2020). This pattern of results resembles Furley et al. (2018) who found that

professional surfers preferred facing the waves in competition like the adaptive/para surfers in this study. There was a significant effect of the total number of frontside waves versus backside waves that contributed to the final score. Facing the wave has shown higher scoring potential for elite surfers possibly due to the ability to spot landings during high-risk maneuvers (Lundgren et al., 2013; Tran et al., 2015). This might contribute to the finding that more frontside waves went into the final score (best two waves of a heat) as frontside surfing is facilitative for the motor response synergies necessary for skillful surfing which are modulated by perceptual information specifying the breaking wave and the spatial orientation of the surfer (Furley et al., 2018; Mendez-Villanueva et al., 2010)

Laterality, goofy versus regular stance, on surfing performance

Lateral preference for a goofy stance or a regular stance demonstrated no significant difference and no effect on surfing performance in surfers with upper limb or lower limb impairments. Most of the participants in this study had a goofy stance ($n = 46$; 55%) over a regular stance ($n = 39$; 45%) which is opposite compared to findings by Furley et al. (2018) of amateur able-bodied surfers (goofy stance = 42%; regular stance = 58%) and professional able-bodied surfers (goofy stance = 36% and regular stance = 64%). Similarly, Chi-square test revealed that the goofy vs. regular distribution were significantly different from the amateur ($\chi^2 (1, n = 478) = 5.016$; $p = .025$) and professional able-bodied surfers ($\chi^2 (1, n = 140) = 4.796$; $p = .029$) with no effect on overall performance (Furley et al., 2018). One rationale for why professional adaptive/para surfers had comparable results as professional able-bodied surfers could be due to deliberate practice and high skill acquisition (Furley & Dörr, 2016; Hambrick et al., 2016). Additionally, the

results are comparable to able-bodied male and female surfers with no found significant difference between goofy and regular. One study found men outperformed women on single judged waves and top two judged wave totals with statistical significance (Barlow et al., 2016). Barlow et al. (2016) analyzed 31 high level competitive women surfers and found their participants had different positioning tactics to catch waves, less aggression with paddling into waves, and different competition strategies, even though they had the same wave counts as their male counterparts.

Use of prosthesis on the front versus the back foot

The third hypothesis was rejected due to no significant difference being found in surfers who used a prosthesis on their front leg versus their back leg. The surfboard is controlled by the back foot and hips with the back foot responsible for turning and maneuvering the board (Mendez-Villanueva et al., 2010). Nathanson (2012) highlighted the importance of maintaining back foot contact with the board, landing on both feet square on the surfboard, and equally absorbing the shock of the landing after high-performance maneuvers, such as an aerial. Although aerial maneuvers were not observed in this study, perhaps because the surfers did not have the opportunity to perform these high-stakes maneuvers during the testing date range, they are assumed to be included within the adaptive/para surfers' technical range of abilities. In addition, Forsyth et al. (2020) found ankle kinematic task requirements that differed from the front foot compared to the back foot using detailed range of motion measurements for both ankles. For this study, wave direction and surf stance were analyzed, and not performance maneuvers. Prosthetic design has improved over the past decade with specific ankle components designed for surfing (Curran & Lyle, 2012); with case studies such as depicting surfing success based on custom-tailored surfing prosthesis (Schvitz et al., 2018); including critical alignment (Burkett et al., 2001). Perhaps improved prosthesis design and surfers' experience contributed to no performance differences of the prosthesis worn on the front or back leg.

Surfers with transfemoral amputations use versus not use of a prosthesis

Surfers with TFA have the choice to surf standing with a prosthesis, surf kneeling without a prosthesis, or position in a deep squatted position using one foot on the surfboard and one hand, in place of the amputated foot. Although the biomechanics of surfing with and without a prosthesis are different, the results indicated that there was no significant difference between the surfers with TFA who wore a prosthesis compared to those who did not. Many individuals with upper and lower limb amputations do not to use a prosthesis in sport due to financial strain, awareness of sport-specific prosthesis, or preference not to use one (Bragaru et al., 2011). These results positively support the global progression of competitive adaptive/para surfing by suggesting surfers without a prosthesis may have similar opportunities and potential for receiving high competition scores and

rankings. Sporting excellence is based on training, not expensive equipment, like a high-tech waterproof prosthesis.

Surfers with upper limb impairments performance based on laterality

No performance difference was evident depending on whether the back, front, or both arms were affected. Hence, the present analysis indicates that upper limb impairments do not differentially impact backside or frontside surfing depending on which limb is affected. However, these results are based on low-powered tests due to limited sample size in the respective categories.

Sport training

Although asymmetries can impair athletic performance, Maloney (2019) found reducing asymmetries through specific sports training can improve overall sports performance. David et al. (2021a) found in 47 competitive adaptive/para surfers, that the number of surf practice sessions and the hours of surf practice were positively correlated with better competition ranking. Additionally, there was no significant correlation with final competition ranking found with any other type of training including strength, cardiovascular, or swimming.

Information on training regimes for adaptive/para surfers is limited in the literature to participants who required physical assistance for surfing or training. David et al. (2023), in a correlational study, found that adaptive/para surfers who required physical assistance ($n = 159$), had fewer number of practice sessions; less hours of surf practice per month; less total surf practice hours per month; decreased years of para surfing experience; and fewer number of surf competitions entered compared with those who did not require physical assistance. Another correlational study of surfers with spinal cord injury ($n = 36$) who needed physical assistance practiced less and ranked lower in competition (David et al., 2021b).

This study was limited to the archived data from classification records, available video footage, and scoring data from contest websites. Future longitudinal studies are warranted to conclude the data with use of video footage analysis of the biomechanics of adaptive surfing and use of prostheses in competition.

The present research approach has some notable strengths and weaknesses. We consider it a strength that the study findings stem from actual in-situ sports competitions, instead of being artificially created laboratory findings. Hence, the findings are likely representative of adaptive/para surfing performance. However, the statistical power of some of the inferential statistical tests conducted was low, and these tests were only able to identify large effects. Therefore, it is possible that small effects remained undetected and future research should scrutinize potential effects resulting from the interaction of laterality, environmental factors, and prosthesis wear in adaptive surfing. Further, the statistical analyses did not account for potential confounding factors like wave conditions or event location, which arguably might have influenced the pattern of results. Although this study acknowledged training and skill acquisition as key factors in performance, inquiries including

direct data on training, volume, and practice structure would be a valuable contribution to the literature on adaptive/para surfing at the competitive level.

Conclusion

This study is the first to investigate performance differences in professional adaptive/para surfers with upper and/or lower limb amputations, congenital limb situations, or dysmelia who surf in the standing position. The impact of surf stance, wave direction, and use of prosthesis was found to have no significant difference although most surfers preferred to surf facing the wave. Adaptive surfing performance may be more influenced by training and skill acquisition than laterality or prosthesis requiring further inquiry.

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